

NVIDIA Earth-2

Open, production-ready AI models and tools for global forecasting and planetary resilience.

Talk summary

NVIDIA Earth-2 is an open, GPU-accelerated platform for building end-to-end AI weather and climate workflows—from assimilating observations and generating atmospheric initial conditions to global medium-range forecasting, storm nowcasting, and regional downscaling. Its models can be run, fine-tuned, and deployed on local infrastructure through Earth2Studio and PhysicsNeMo, complementing traditional numerical weather prediction with substantially faster and more energy-efficient inference. For the weather and climate community, Earth-2's main contribution is a composable AI pipeline that connects global prediction with kilometer-scale local intelligence while making advanced forecasting capabilities more accessible to researchers, operational agencies, and climate-risk applications.

AI models featured in Earth-2

- **Earth-2 Medium Range (Atlas):** Predicts more than 70 atmospheric variables—including temperature, pressure, wind, and humidity—up to 15 days ahead.
- **Earth-2 Nowcasting (StormScope):** Uses generative AI and satellite/radar imagery to produce kilometer-resolution forecasts of storms and hazardous weather from zero to six hours ahead.
- **Earth-2 Global Data Assimilation (HealDA):** Converts observations from satellites, weather balloons, and surface stations into global atmospheric initial conditions in seconds on GPUs.
- **Earth-2 CorrDiff:** Generatively downscales coarse continental forecasts into high-resolution regional weather fields, reportedly up to 500 times faster than traditional methods.

- **Earth-2 FourCastNet 3:** Provides accelerated global forecasts for variables such as wind, temperature, and humidity, with particular emphasis on scalable datasets and extreme-weather prediction.